

Report for dataset 20241115_Liverpool_202345_Cell1-Cell1

Dataset 0628c687-eb37-45b4-ac73-640ab3c2d933

Summary

Name	20241115_Liverpool_202345_Cell1-Cell1
Created At	2024-11-16 22:25:34.502
Number of Records	5,570,543
Total Length	84,082,791,746
Movie Name	m84105_241115_114711_s4
ICS Version	13.1.0.221972
Well Sample	20241115_Liverpool_202345_Cell1
Biological Sample	20241115_Liverpool_202345_Cell1

CCS Analysis Report

Summary

HiFi reads	5.6 M
HiFi reads yield	84.00 Gb
HiFi reads length (mean)	15.08 kb
HiFi reads length (median, bp)	14,392
HiFi Read Length N50 (bp)	15,432
HiFi Read Quality (median)	Q35
HiFi Read Quality (median)	35
Base Quality Q30 (%)	93.63%
HiFi Number of Passes (mean)	9

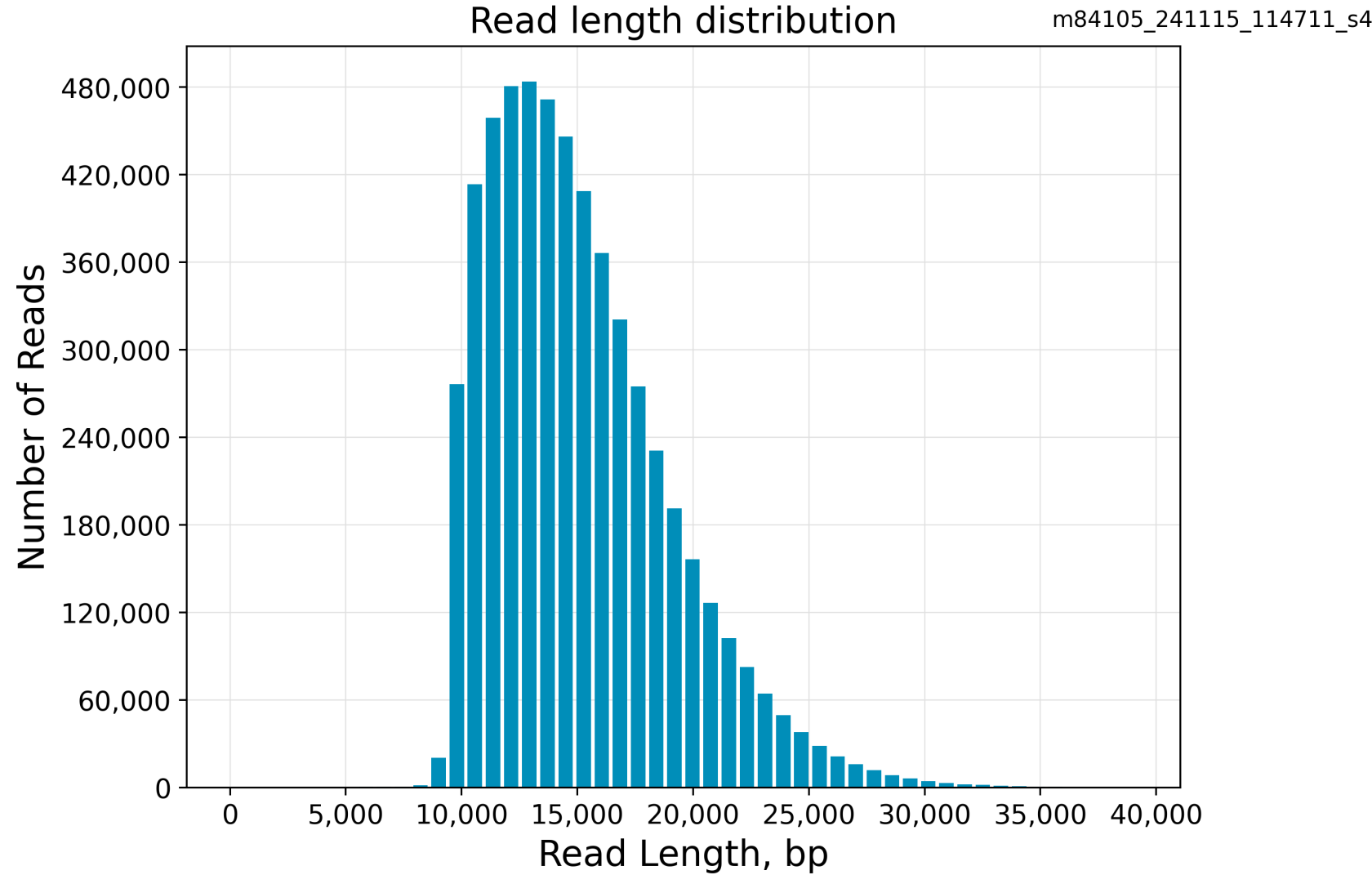
HiFi Read Length Summary

Read Length (kb)	Reads	Reads (%)	Yield (Gb)	Yield (%)
0	5570543	100	84.00 Gb	100
5,000	5570095	100	84.00 Gb	100
10,000	5350725	96	81.88 Gb	97
15,000	2446389	44	45.47 Gb	54
20,000	634937	11	14.50 Gb	17
25,000	108450	2	2.97 Gb	4
30,000	12998	0	0.42 Gb	0
35,000	1257	0	0.05 Gb	0
40,000	133	0	0.01 Gb	0

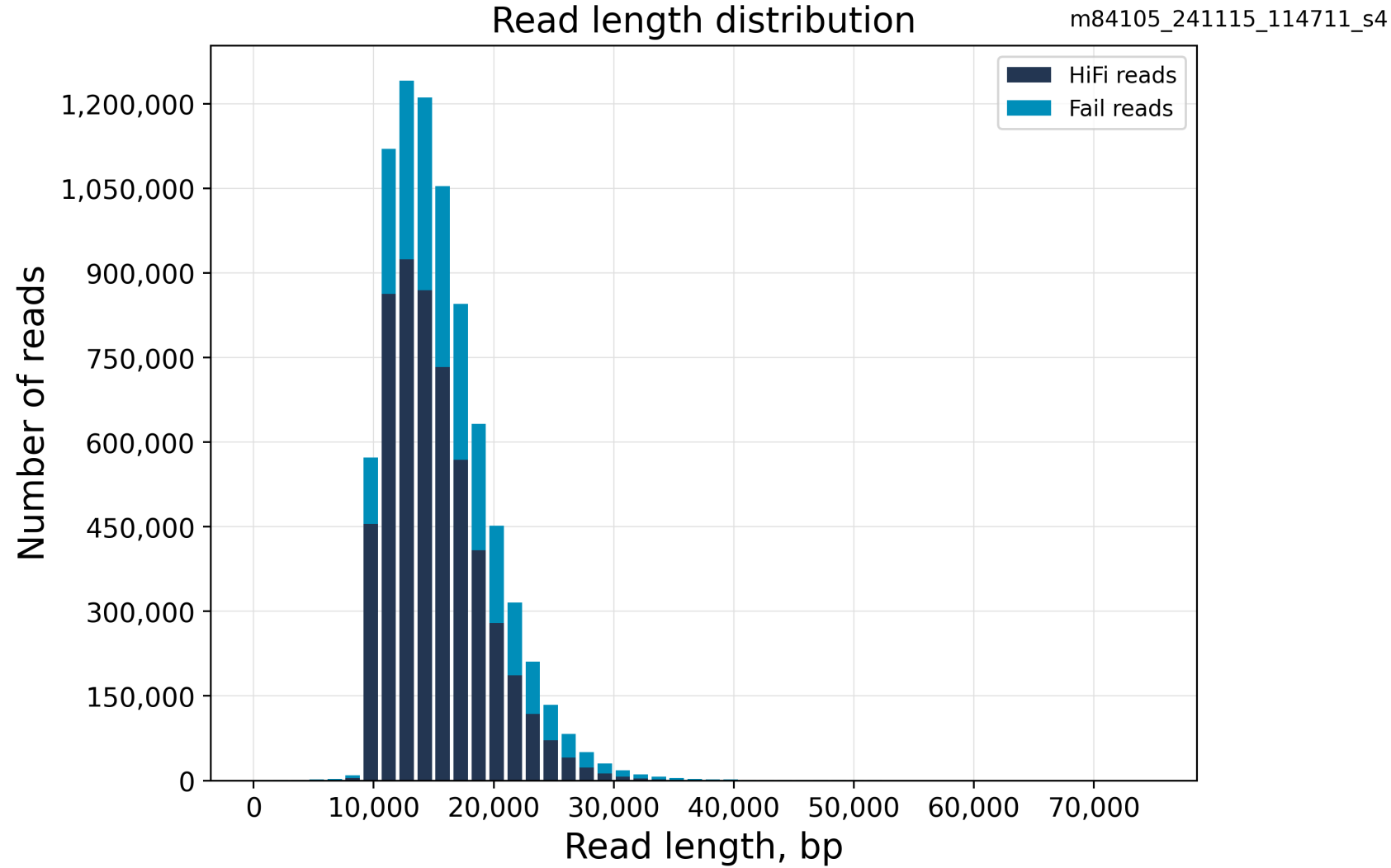
HiFi Read Quality Summary

Read Quality (Phred)	Reads	Reads (%)	Yield (Gb)	Yield (%)
Q20	5570543	100	84.00 Gb	100
Q30	3963975	71	58.21 Gb	69
Q40	1624891	29	21.63 Gb	26
Q50	190996	3	2.19 Gb	3

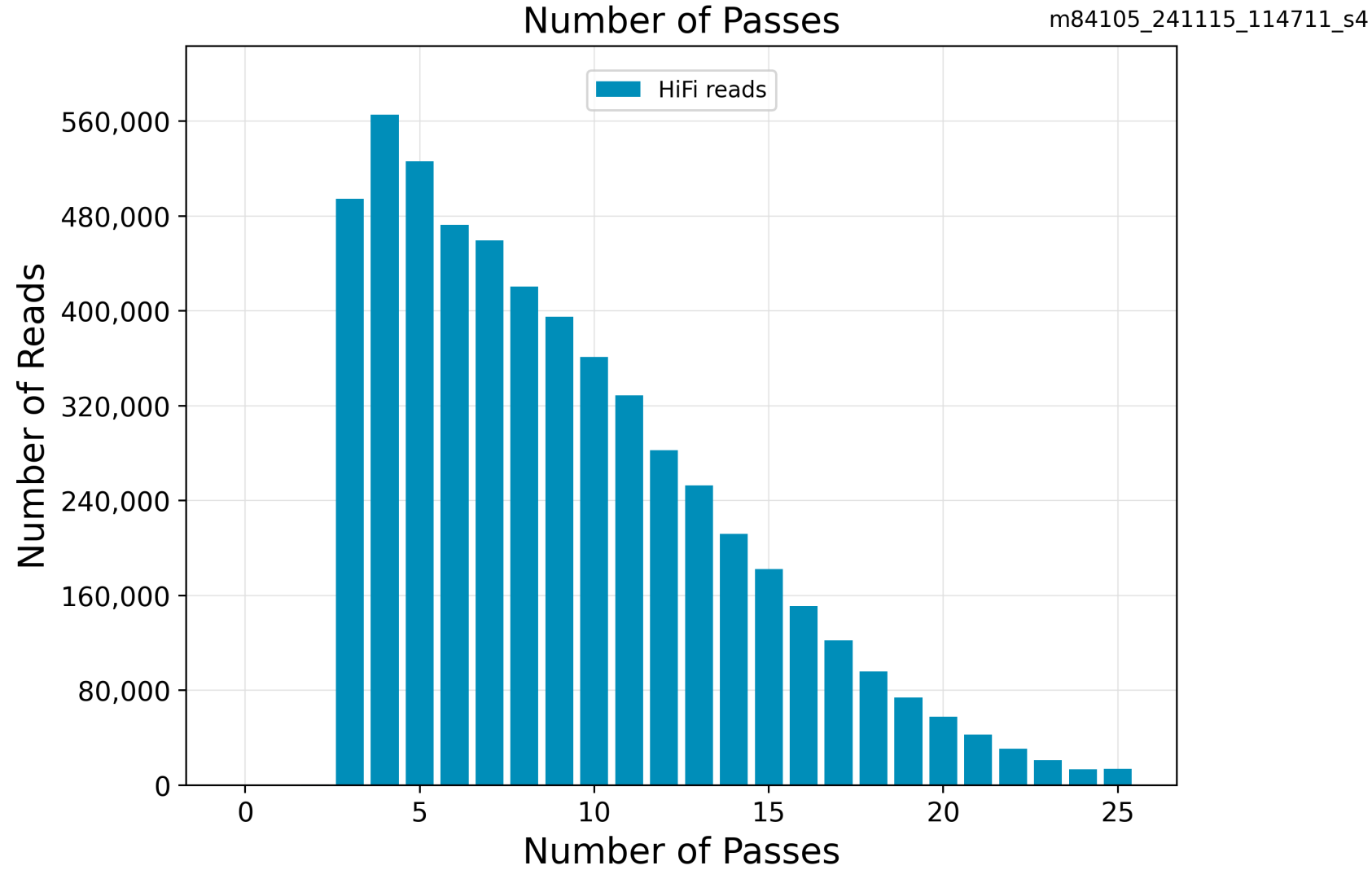
Read length distribution



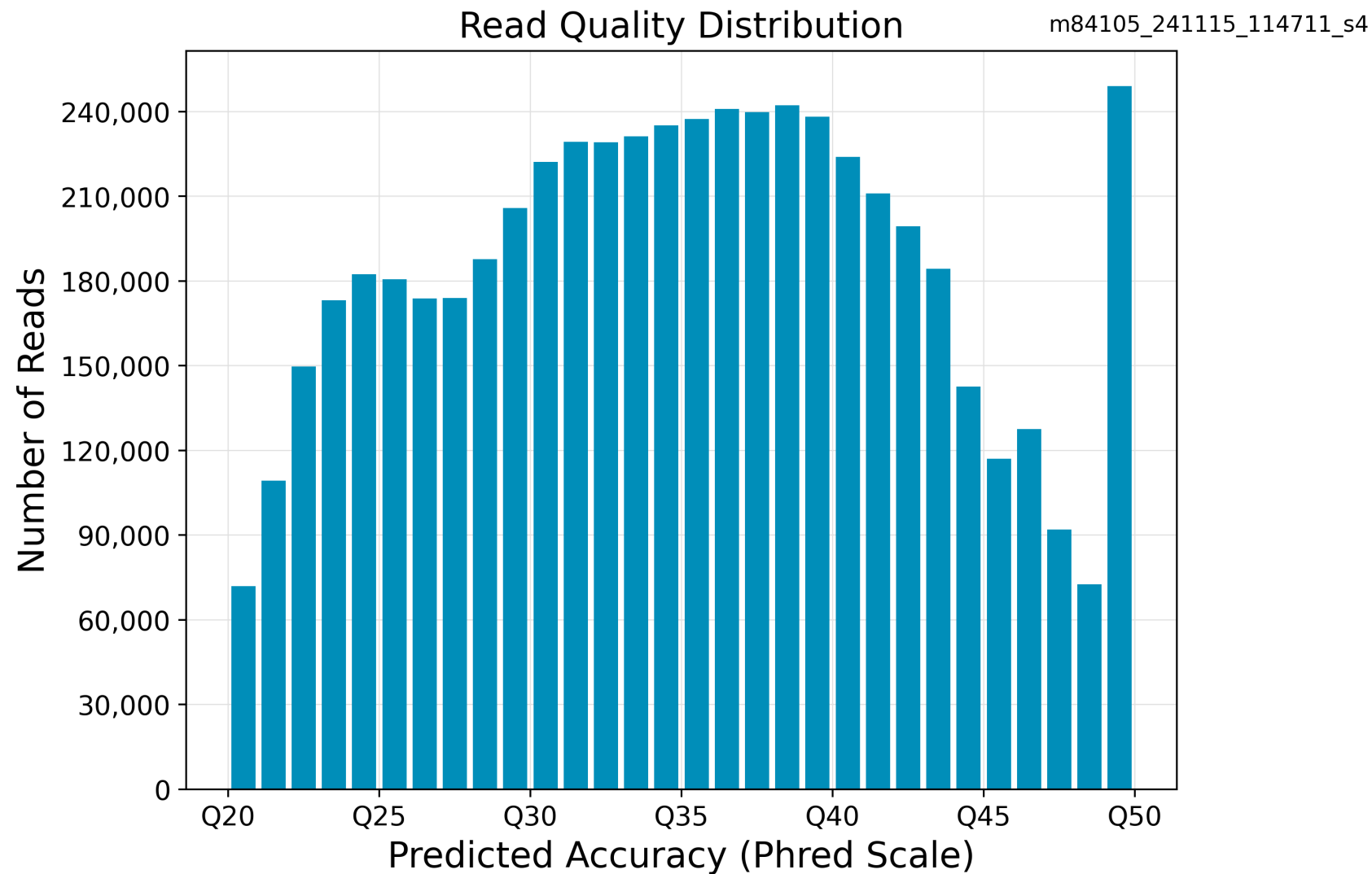
Read length distribution



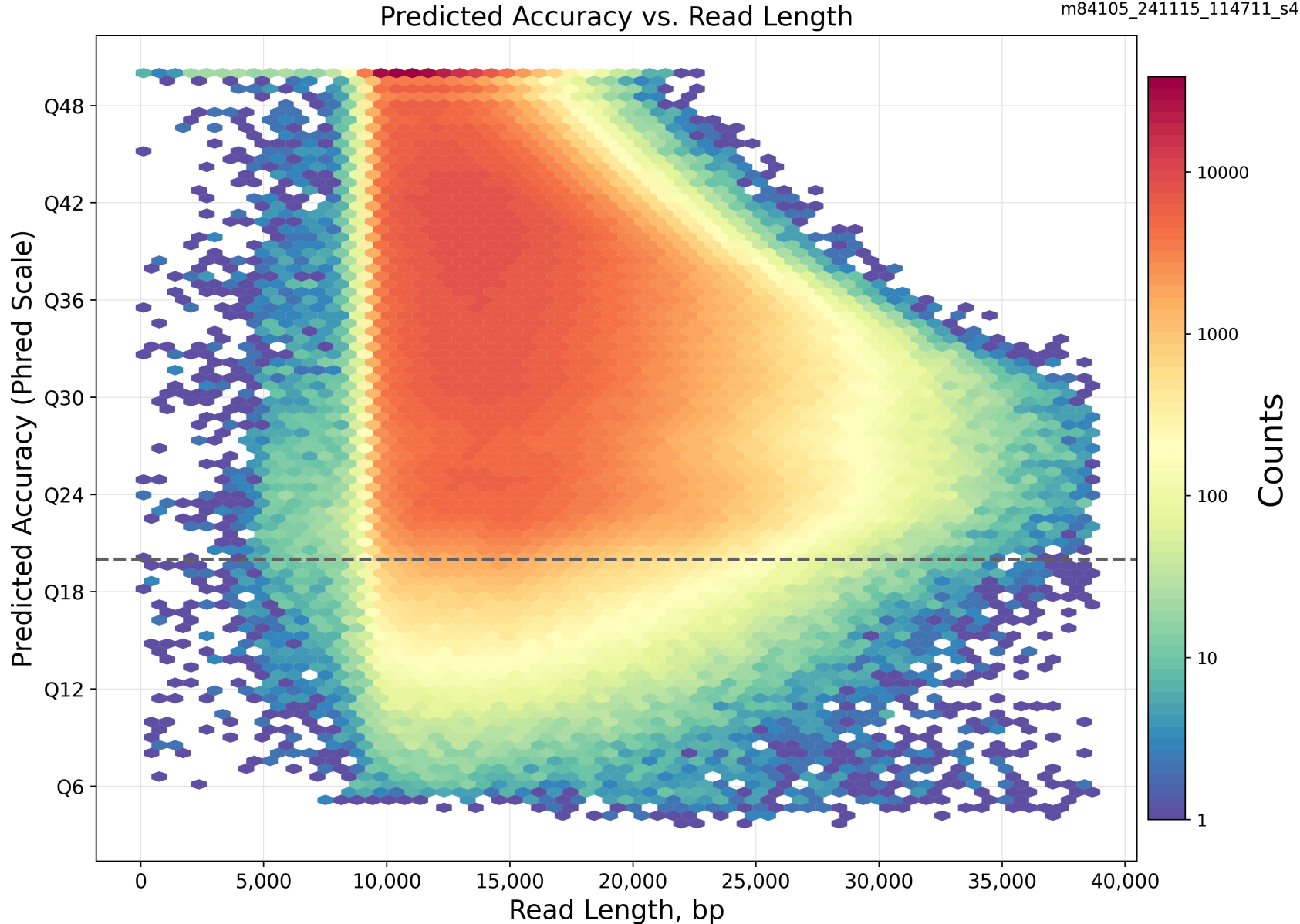
Number of Passes



Read Quality Distribution



Predicted Accuracy vs. Read Length



Adapter Report

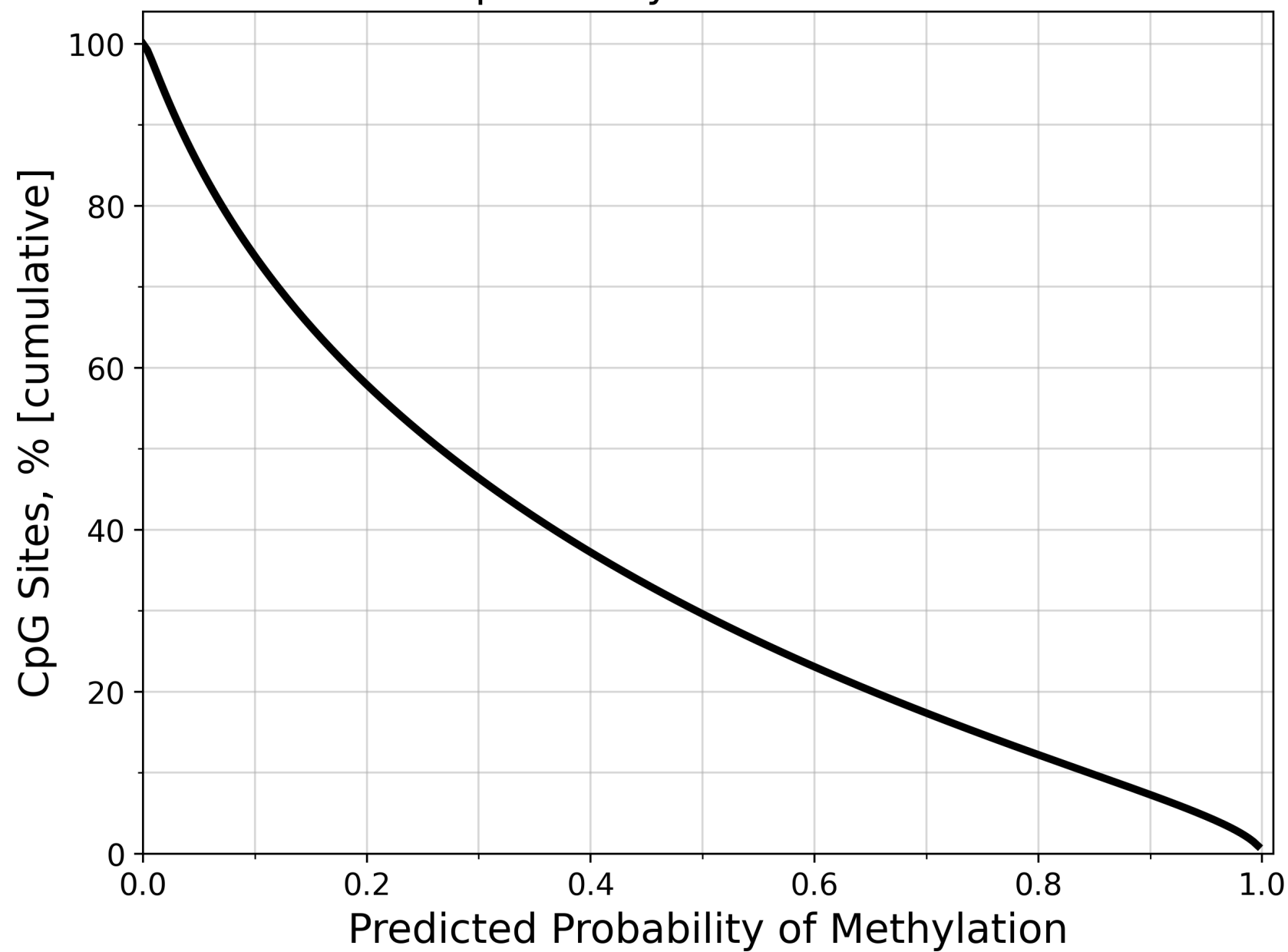
Summary

Adapter Dimers (0-10bp) %	0
Short Inserts (11-100bp) %	0
Local Base Rate	2.86

5mC CpG Report

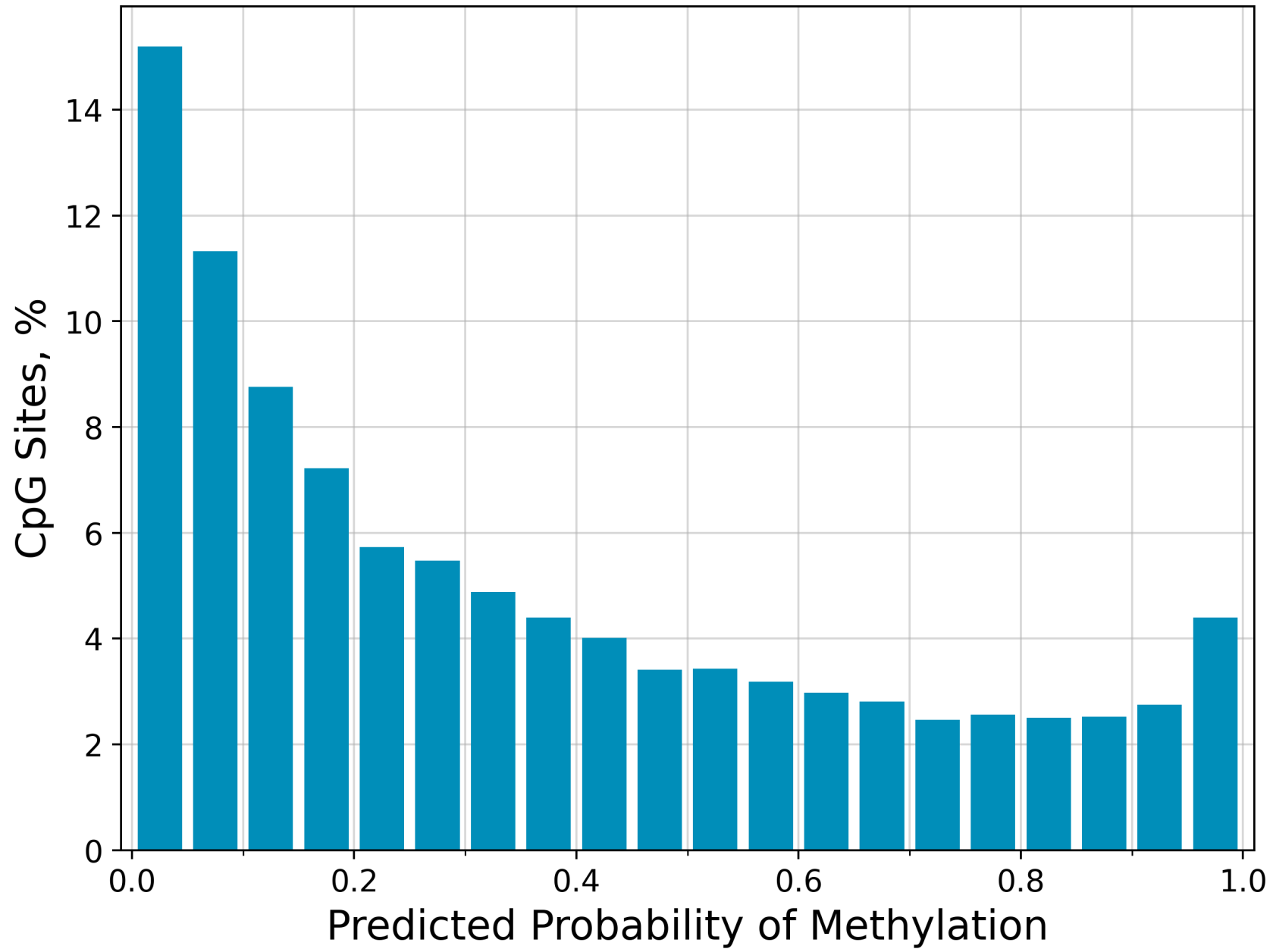
CpG Methylation in Reads

CpG Methylation in Reads



CpG Methylation in Reads (Histogram)

CpG Methylation in Reads (Histogram)

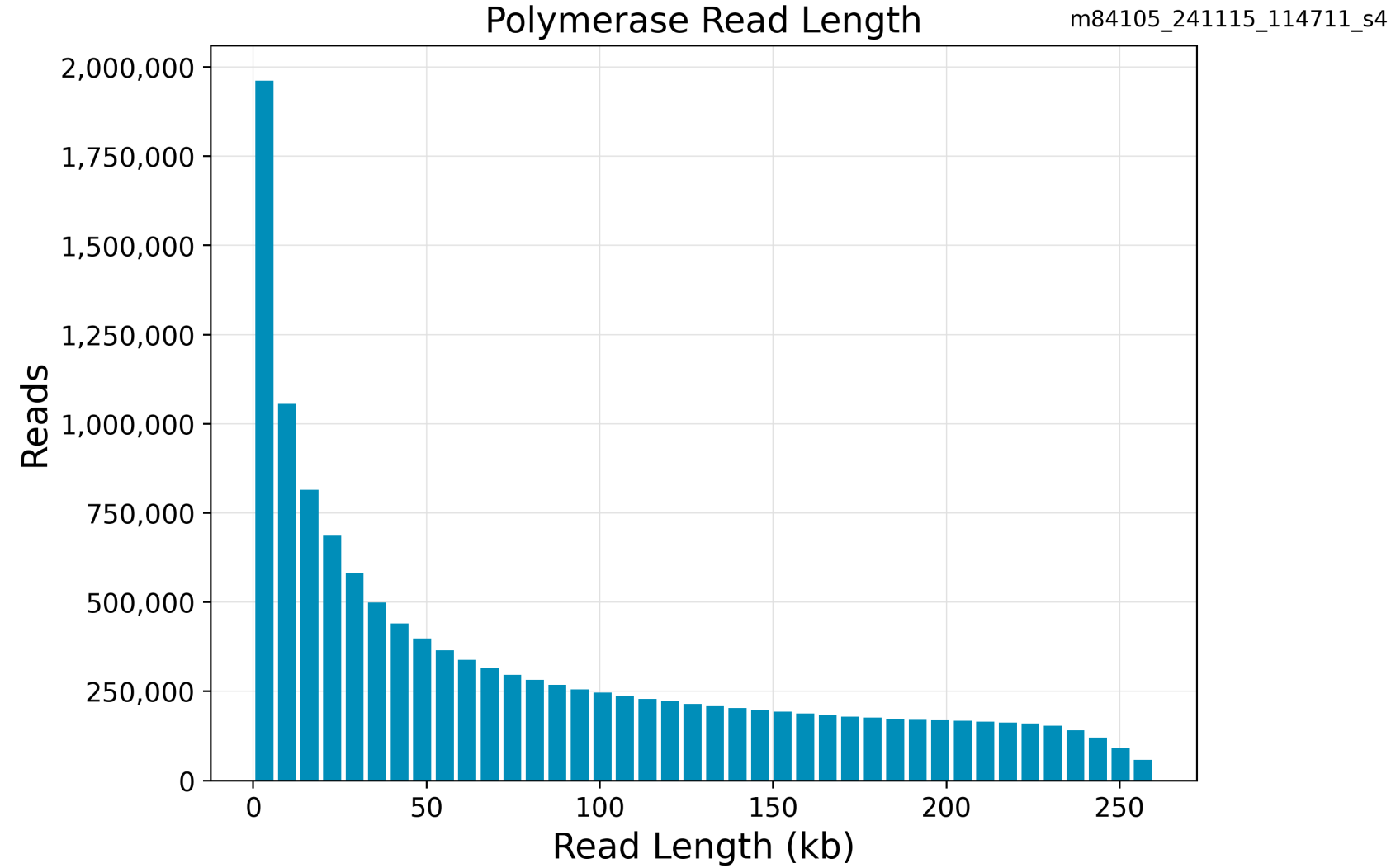


Raw Data Report

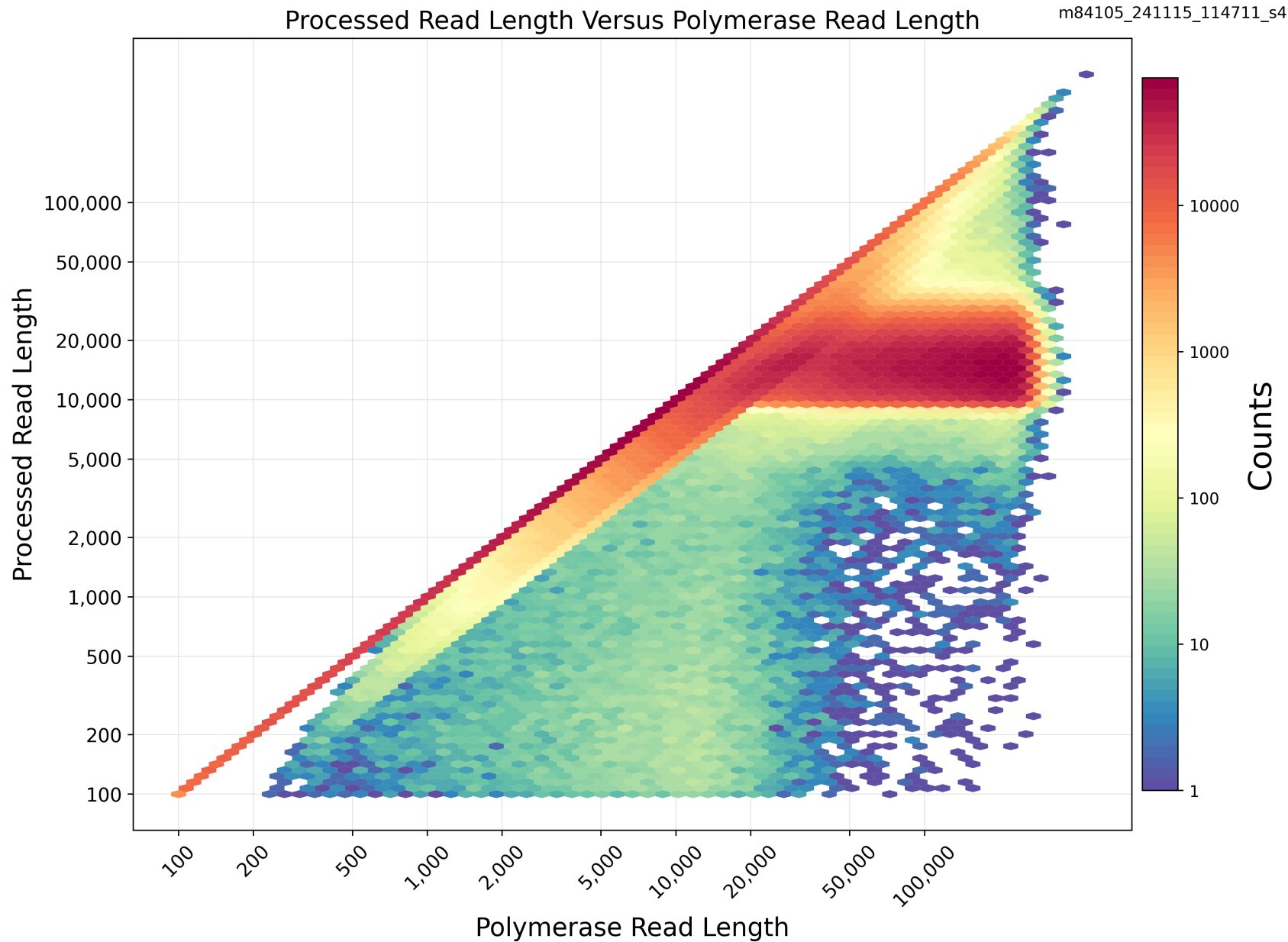
Summary

Polymerase Read Bases	1,017,211,948,947
Polymerase Reads	13,053,764
Polymerase Read Length (mean)	77.93 kb
Polymerase read length (N50)	153.75 kb
Polymerase read length longest subread length (mean)	15.52 kb
Polymerase read length longest subread length (N50)	17.75 kb
Unique Molecular Yield	191,736,496,128

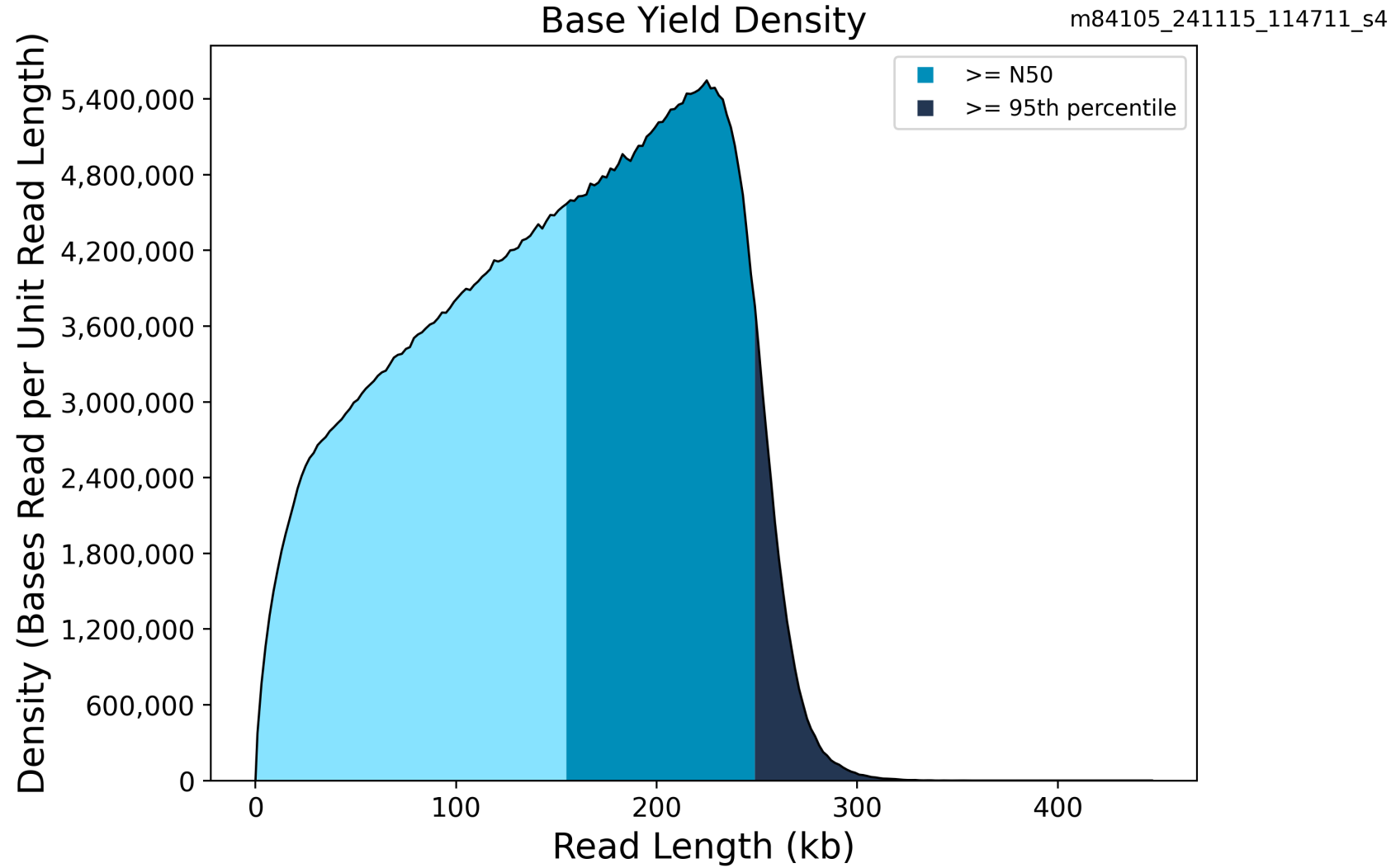
Polymerase Read Length



Longest Subread Length Versus Polymerase Read Length



Base Yield Density



CCS Processing

Summary

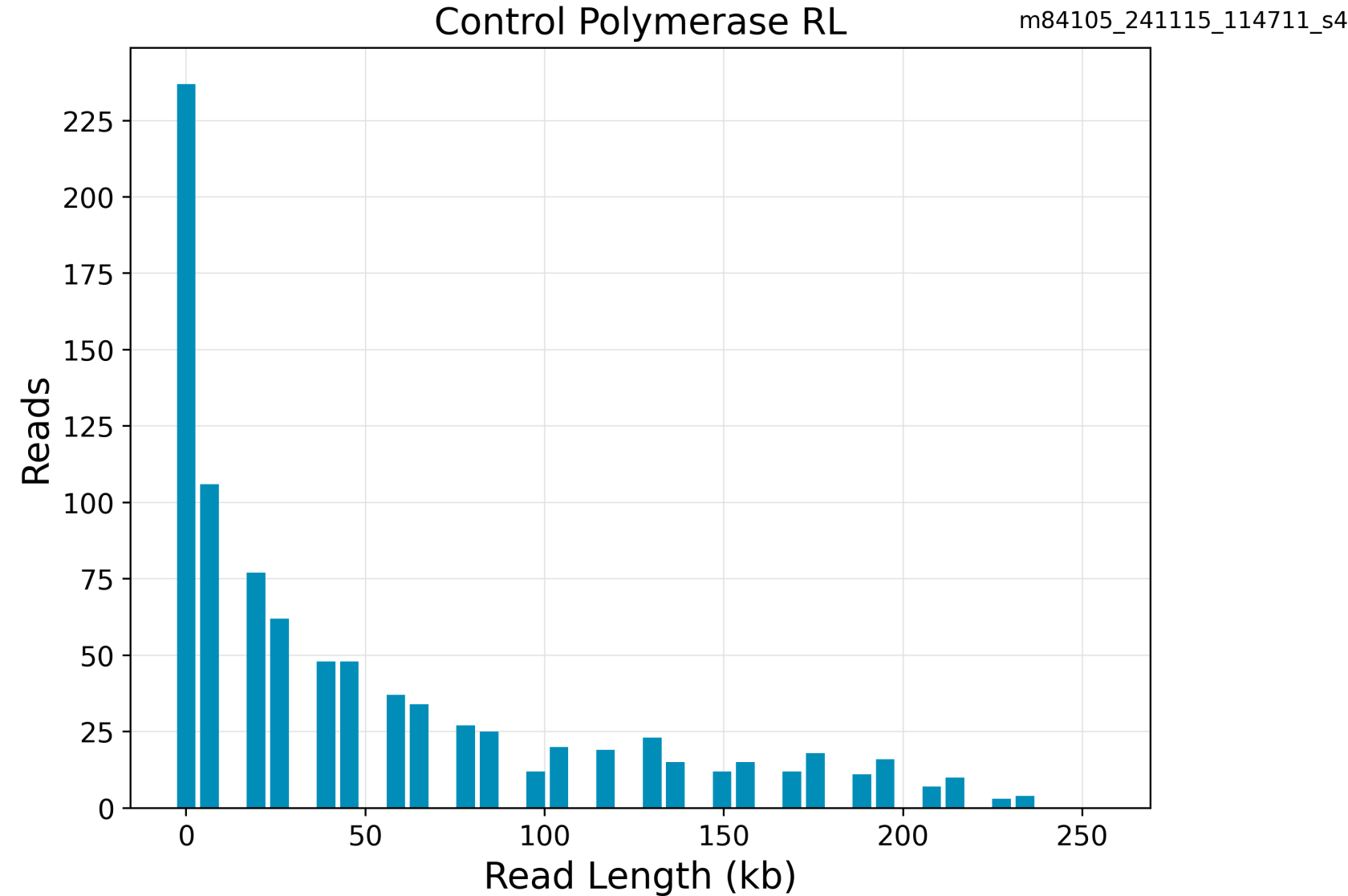
ZMWs input	13,948,905
ZMWs pass filters	5,721,560
ZMWs fail filters	8,227,345
ZMWs shortcut filters	0
ZMWs with tandem repeats	75,348
Below SNR threshold	266,199
Median length filter	953,110
Lacking full passes	4,678,763
Heteroduplex insertions	0
Coverage drops	0
Insufficient draft cov	0
Draft too different	0
Draft generation error	0
Draft above --max-length	0
Draft below --min-length	0
Reads failed polishing	4,168
Empty coverage windows	7,537
CCS did not converge	0
CCS adapter concatenation	0
CCS adapter palindrome	3
CCS adapter residue	20
ZMW with full-length subread	2,316,647
ZMW with control failure	783
ZMW with control success	115
CCS below minimum RQ	0
Unknown error	0

ZMWs missing adapters	80,984
HiFi Reads	5,570,543
HiFi Yield (bp)	84,082,791,746
HiFi Read Length (mean, bp)	15,094
HiFi Read Length (median, bp)	14,406
HiFi Read Length N50 (bp)	14,406
HiFi Read Quality (median)	35
HiFi Number of Passes (mean)	9
<Q20 Reads	151,017
<Q20 Yield (bp)	2,326,089,237
<Q20 Read Length (mean, bp)	15,402
<Q20 Read Length (median, bp)	14,711
<Q20 Read Quality (median)	18
>=Q30 Reads	3,963,959
>=Q30 Yield (bp)	58,265,058,773
>=Q30 Read Length (mean, bp)	14,698
>=Q30 Read Length (median, bp)	14,054
>=Q30 Read Quality (median)	38
Base quality >=Q30 (bp)	78,730,344,738
Base quality >=Q30 (%)	93.60000000000001

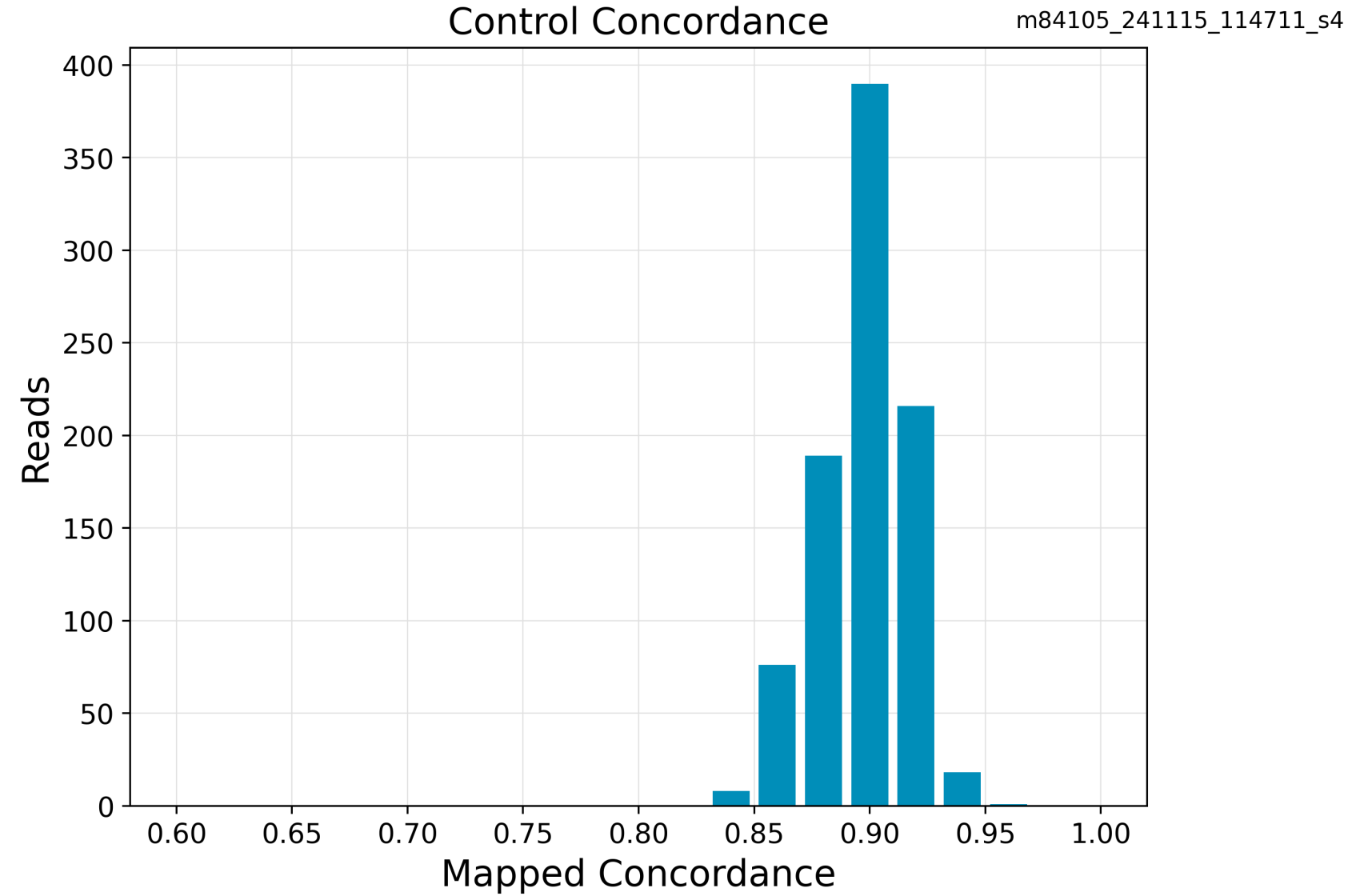
Control Report

Summary

Number of Control Reads	898
Control Read Length Mean	54,566
Control Read Concordance Mean	0.91
Control Read Concordance Mode	0.91



Control Concordance



Run Preview

Summary

Estimated HiFi Yield (Gb)	None
HiFi read quality, median	None
HiFi read length, mean	14,279
P1 (%)	52
Time (Hrs)	4
Workflow	Basic
Movie Name	m84105_241115_114711_s4_basic_preview_4hr
Movie Id	m84105_241115_114711_s4

Barcode Preview

Barcode	HiFi reads	HiFi read length (mean, bp)
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Run Preview

Summary

Estimated HiFi Yield (Gb)	80,008,155,755
HiFi read quality, median	34
HiFi read length, mean	15,017
P1 (%)	None
Time (Hrs)	23
Workflow	Full
Movie Name	m84105_241115_114711_s4_full_preview_23hr
Movie Id	m84105_241115_114711_s4

Barcode Preview

Barcode	HiFi reads	HiFi read length (mean, bp)
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Loading Report

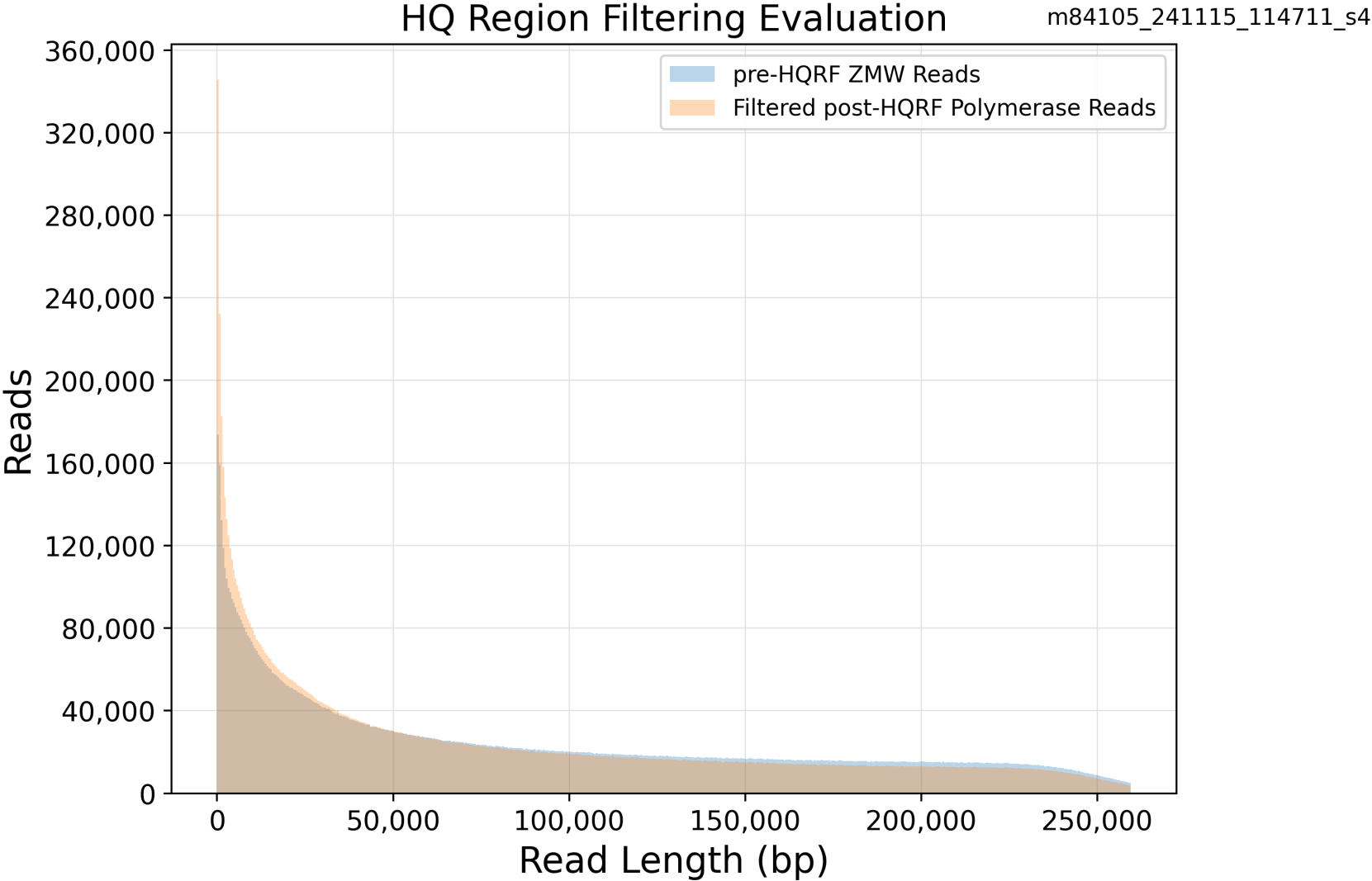
Summary

Productive ZMWs	25,165,824
Productivity 0	11,994,931
Productivity 1	13,054,662
Productivity 2	116,231

Loading Statistics

Collection Context	Total ZMWs	Productivity 0	P0 (%)	Productivity 1	P1 (%)	Productivity 2	P2 (%)	Loading type
m84105_241115_114711_s4	25165824	11994931	47.66	13054662	51.88	116231	0.46	Workflow_Kestrel.py

HQ Region Filtering Evaluation



No Sample Setup found

Instrument run(s)

Run d1c79938-1ff4-4793-92e8-00e68e17a29a

Summary

Name	20241115_Liverpool
Status	COMPLETE
Created	2024-11-15 11:25:11.016
Started	2024-11-15 11:40:08.591
Completed	2024-11-16 22:01:11.298
Context	r84105_20241115_113945
Instrument Name	Revio
Instrument Serial Number	84105
ICS Version	13.1.0.221972
Primary Analysis Version	13.1.0.221972
Chemistry Version	13.1.0.217683

Parent jobs (1)

Job 8763

Summary

Job Type	import-dataset
Name	import-dataset
Comments	Description for job Import PacBio DataSet
Created At	2024-11-16 23:17:20.614
SMRT Link Version	13.1.0.221970

Child jobs (1)

Job 8768

Summary

Job Type	analysis
Pipeline	cromwell.workflows.pb_export_ccs
Name	20241119_Liverpool_Fastq - 20241115_Liverpool_202345_Cell1-Cell1
Comments	Deferred Job 20241119_Liverpool_Fastq - 20241115_Liverpool_202345_Cell1-Cell1
Created At	2024-11-19 16:01:48.530
SMRT Link Version	13.1.0.221970

Output files

File name	Size	Type
m84105_241115_114711_s4_fastq.zip	32,725,858,004	zip
m84105_241115_114711_s4_exported.bam	44,565,475,921	bam
Analysis Log	13,112	log
SMRT Link Log	9,545	log

